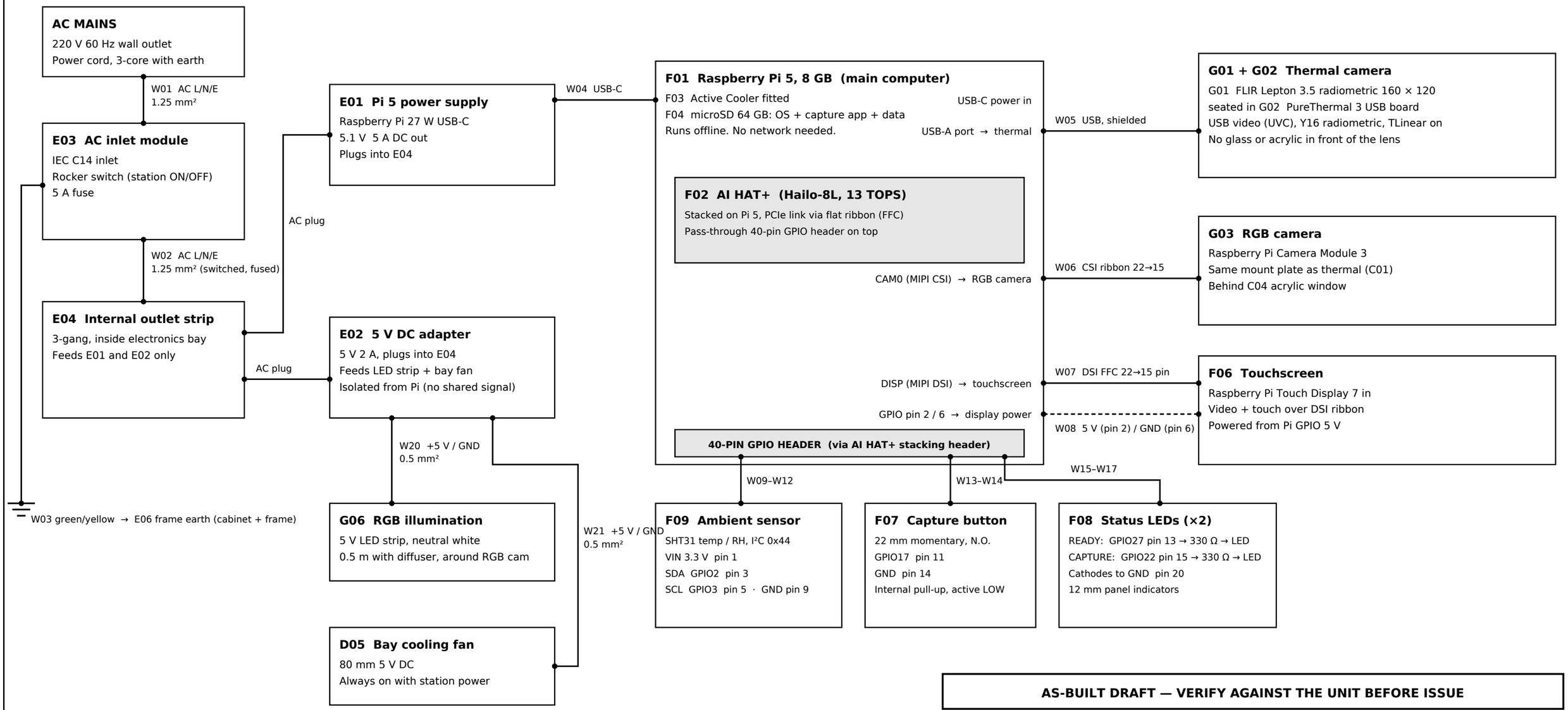
ELECTRICAL WIRING DIAGRAM — DFUSCAN-01 FOOT THERMAL STATION

Block-level wiring of every powered part. Part numbers match the BOM (DFUSCAN-xNN). Wire numbers match the wire schedule on sheet 2.



AS-BUILT DRAFT — VERIFY AGAINST THE UNIT BEFORE ISSUE

eMathrix Technologies	DFUSCAN-01 · IR Thermography Foot Station
Sheet	Electrical wiring diagram (ELEC-03)
Drawing no.	**EMX-2026-DFUSCAN-01-ELEC-03**
Rev	A · Oct 27, 2026 · first issue, as-built draft
Scale / units	NTS · not to scale · electrical only
Drawn / checked / approved	eMathrix draft · PM: _____ · Owner: _____
Sheet	1 of 2

NOTES

1. The 40-pin header is reached through the AI HAT+ stacking header. Pin numbers are physical (board) numbers; GPIO numbers are BCM.
2. Logic on the GPIO header is 3.3 V. Nothing at 5 V may connect to a GPIO pin. F09 runs on 3.3 V (pin 1).
3. E02 (LED strip, fan) is a separate 5 V supply with no signal link to the Pi, so it needs no common ground.
4. Every AC metal part (cabinet, frame) is bonded to the inlet earth (W03). The AC section stays inside the closed bay.
5. Pin assignments shown are the design intent. The PM confirms each one against the built unit before this sheet is issued.
6. Thermal camera optics: the Lepton looks through an open port. Glass and acrylic block long-wave IR.

PIN MAP AND WIRE SCHEDULE — DFUSCAN-01

ELEC-04 pin map (handover sheet) and wire schedule for every connection on sheet 1.

ELEC-04 PIN MAP (Raspberry Pi 5 + AI HAT+)

Phys. pin	BCM GPIO	Connected to	Signal	Voltage	Pull	Software name
1	3V3	F09 SHT31 VIN	Sensor supply	3.3 V	—	—
2	5V	F06 touchscreen 5 V	Display supply	5 V	—	—
3	GPIO2 (SDA1)	F09 SHT31 SDA	I²C data	3.3 V	Board pull-up	i2c-1 / 0x44
5	GPIO3 (SCL1)	F09 SHT31 SCL	I²C clock	3.3 V	Board pull-up	i2c-1 / 0x44
6	GND	F06 touchscreen GND	Ground	0 V	—	—
9	GND	F09 SHT31 GND	Ground	0 V	—	—
11	GPIO17	F07 capture button	Digital in, active LOW	3.3 V	Internal pull-up	BTN_CAPTURE
13	GPIO27	F08 READY LED (via 330 Ω)	Digital out	3.3 V	—	LED_READY
14	GND	F07 capture button	Ground	0 V	—	—
15	GPIO22	F08 CAPTURE LED (via 330 Ω)	Digital out	3.3 V	—	LED_CAPTURE
20	GND	F08 LED cathodes	Ground	0 V	—	—
—	USB-A	G02 PureThermal 3 (G01 Lepton 3.5)	USB UVC video, Y16	5 V bus	—	/dev/video (thermal)
—	CAM0	G03 Camera Module 3	MIPI CSI-2	—	—	Picamera2 camera 0
—	DISP	F06 touchscreen	MIPI DSI + touch	—	—	display 0
—	PCIe FFC	F02 AI HAT+ (Hailo-8L)	PCIe Gen 3 ×1	—	—	HailoRT device

WIRE SCHEDULE

Wire	From	To	Function	Type / gauge	Colour
W01	Wall outlet	E03 AC inlet	AC L / N / E	Power cord 3 × 1.25 mm²	Brown / blue / green-yellow
W02	E03 inlet (switched, fused)	E04 outlet strip	AC L / N / E	1.25 mm² stranded	Black / white / green-yellow
W03	E03 inlet earth	Frame + cabinet (E06)	Protective earth	1.25 mm², ring lugs	Green-yellow
W04	E01 PSU	Pi 5 USB-C	5.1 V 5 A	Captive USB-C lead	—
W05	Pi 5 USB-A	G02 PureThermal 3	USB data + 5 V	USB-C to USB-A, shielded, short	—
W06	Pi 5 CAM0	G03 Camera Module 3	CSI	FFC 22 → 15 pin, 300 mm	—
W07	Pi 5 DISP	F06 touchscreen	DSI	FFC 22 → 15 pin	—
W08	GPIO pin 2 / 6	F06 touchscreen	5 V / GND	22 AWG jumper pair	Red / black
W09–W12	GPIO pins 1, 3, 5, 9	F09 SHT31	3.3 V / SDA / SCL / GND	22 AWG, 4-core	Orange / blue / yellow / black
W13–W14	GPIO pins 11, 14	F07 button	Signal / GND	22 AWG pair	White / black
W15–W17	GPIO pins 13, 15, 20	F08 LEDs (330 Ω inline)	READY / CAPTURE / GND	22 AWG, 3-core, heat shrink	Green / red / black
W20	E02 adapter	G06 LED strip	+5 V / GND	0.5 mm² (20 AWG)	Red / black
W21	E02 adapter	D05 bay fan	+5 V / GND	0.5 mm² (20 AWG)	Red / black

POWER BUDGET (5 V side)

Load	Supply	Typical draw
Pi 5 + AI HAT+ + cooler, running CNN	E01	up to about 12 W
Lepton 3.5 + PureThermal 3	E01 via USB	about 1 W
Camera Module 3	E01 via CSI	under 1 W
7 in touchscreen	E01 via GPIO 5 V	about 2.5 W
SHT31, button, LEDs	E01 via GPIO	negligible
E01 total vs rating	27 W PSU	about 17 W, under 65%
LED strip + fan	E02 (10 W)	about 3.5 W, under 40%

Figures are typical ratings for sizing, not measured on this unit.

Derating target 70–80% met on both supplies.

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Sheet	Pin map and wire schedule (ELEC-04)
Drawing no.	**EMX-2026-DFUSCAN-01-ELEC-04**
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Sheet	2 of 2